Satellite Ephemeris and Coordinates

José E. Sanguino

IT – Instituto de Telecomunicações
Instituto Superior Técnico, Av. Rovisco Pais 1, 1049-001 Lisboa, Portugal
Email: jose.sanguino@tecnico.ulisboa.pt

Planetary Motion and The Kepler's Laws

Johannes Kepler 1571-1630

1. The orbits of the planets are ellipses with the Sun at one *focus* (1602).

2. A vector from the Sun to the planet sweeps out equal areas in equal times (the law of areas, 1605).

3. The square of a planet's orbital period is proportional to the cube of the semi-major axis of the planet's orbit (1618).

The Satellite Orbital Period and Kepler's 3rd Law

Orbital period:

$$T^2 = \frac{4\pi^2}{\mu} A^3 \quad \Longleftrightarrow \quad \boxed{T = 2\pi\sqrt{\frac{A^3}{\mu}}}$$

Earth gravitational constant: $\mu = 3.986005 \times 10^{14}\ m^3 s^{-2}$

(IS-GPS-200)

Example

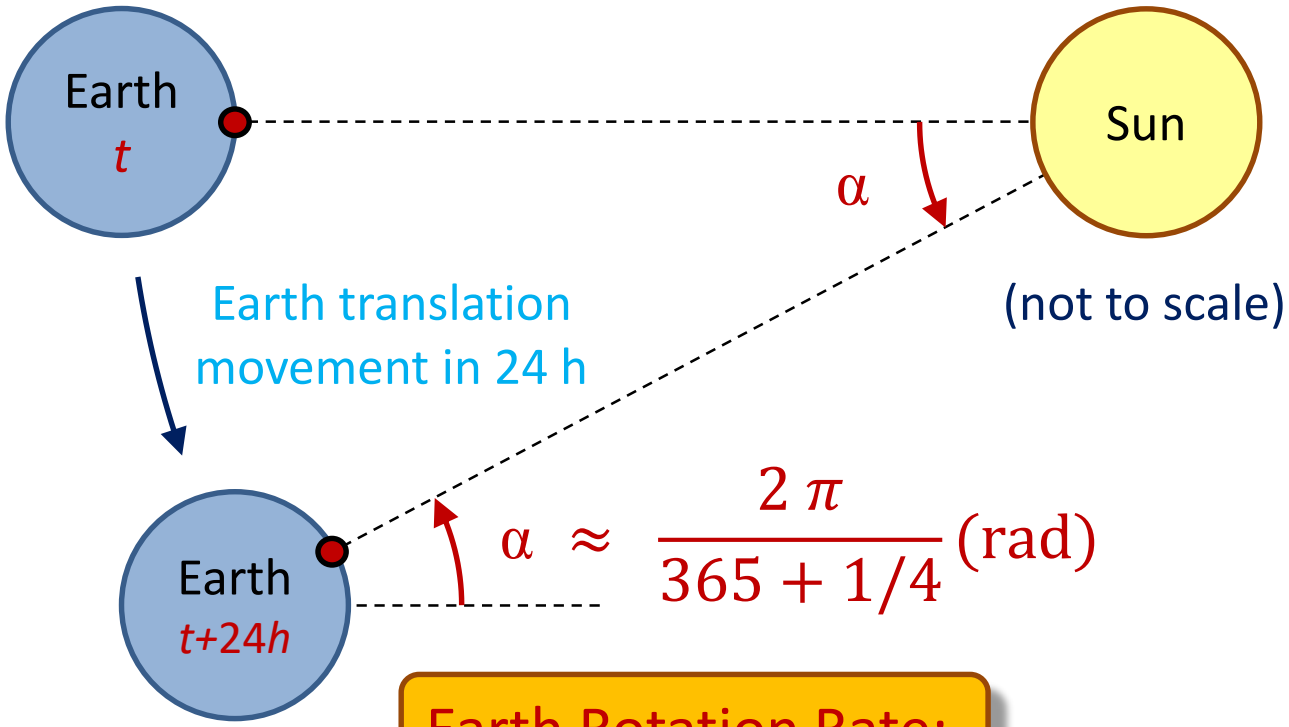
IRIDIUM 33 A = 7160399 m T = 6030 s = 101 minutes

Kepler's 3rd Law

The square of a planet's orbital period is proportional to the cube of the semi-major axis of the planet's orbit (1618).

Earth Rotation Rate

$\dot{\Omega}_e$



$$\alpha \approx \frac{2\,\pi}{365 + 1/4}\,(\mathrm{rad})$$

Earth Rotation Rate:

$$\dot{\Omega}_e \approx \frac{2\pi + \alpha}{24 \times 60 \times 60}\ (\mathrm{rad/s})$$

IS-GPS-200:

$$\dot{\Omega}_e = 7.2921151467 \times 10^{-5}\ \mathrm{rad/s}$$

Geostationary Orbit

Satellite Angular Velocity (ω_s)=
= Earth Rotation Rate ($\dot{\Omega}_e$)

$$\omega_s = \dot{\Omega}_e = 7.2921151467 \times 10^{-5} \text{rad/s}$$

Orbital Period (T)

$$T = \frac{2\pi}{\omega_s} \text{ (s)} = 86164 \text{ s}$$
$$= 23.93 \text{ hour}$$

Orbit Semi-Major Axis (A)

$$A = \sqrt[3]{\frac{\mu\, T^2}{4\,\pi^2}} \text{ (m)}$$
$$= 42164 \text{ km}$$

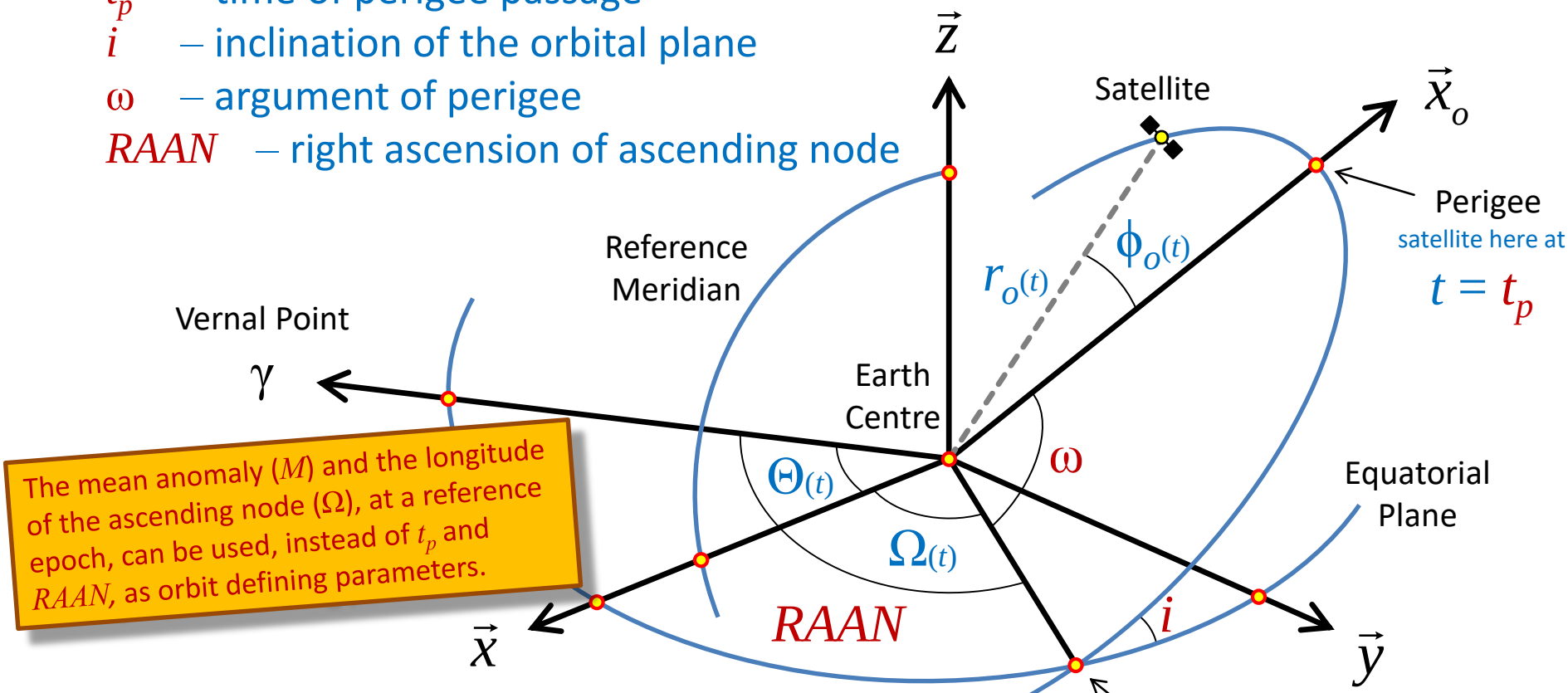
Earth gravitational constant:

$\mu = 3.986005 \times 10^{14}\ \text{m}^3\text{s}^{-2}$

(IS-GPS-200)

Keplerian Orbital Parameters

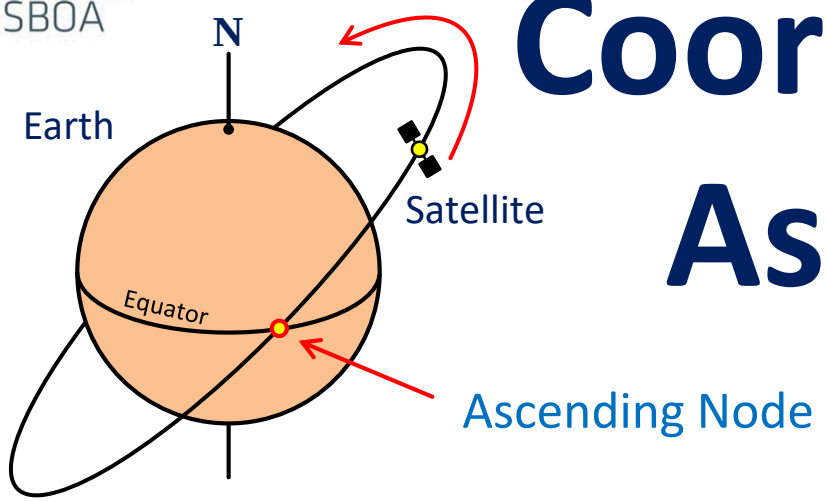
A – semi-major axis of the orbital ellipse
e – eccentricity of the orbital ellipse
t_p – time of perigee passage
i – inclination of the orbital plane
ω – argument of perigee
$RAAN$ – right ascension of ascending node

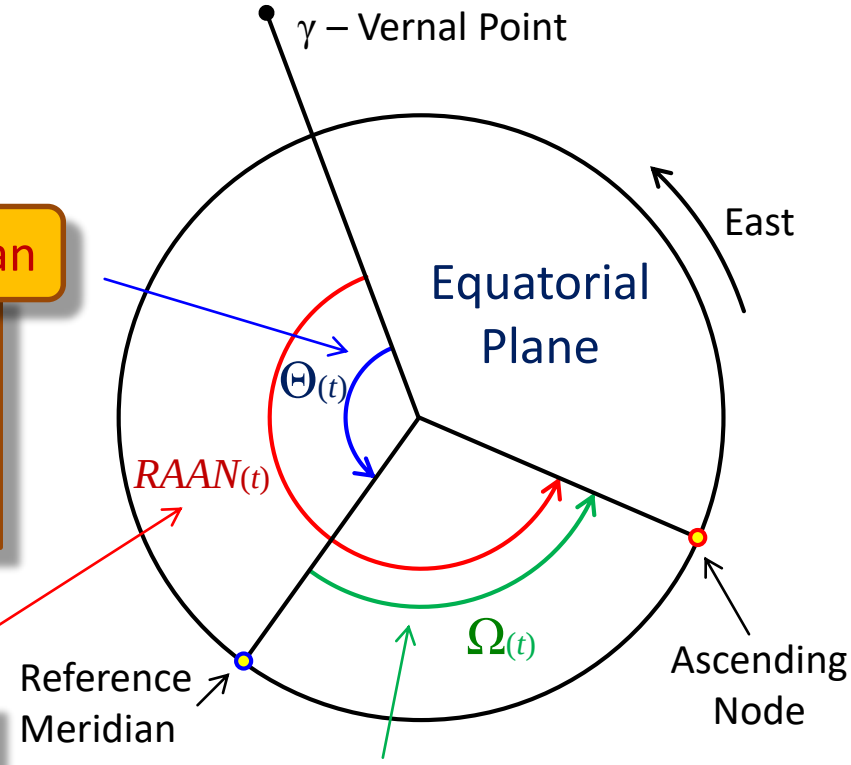


The mean anomaly (M) and the longitude of the ascending node (Ω), at a reference epoch, can be used, instead of t_p and $RAAN$, as orbit defining parameters.

Ω – longitude of the ascending node
Θ – right ascension of the reference meridian

Coordinates of the Ascending Node





Right Ascension of the Reference Meridian

$$\Theta(t) = \Theta(t_o) + \dot{\Omega}_e (t - t_o)$$

Earth rotation rate:

$$\dot{\Omega}_e = 7.2921151467 \times 10^{-5} \, \mathrm{rad\,s^{-1}}$$

Right Ascension of the Ascending Node

$$RAAN(t) = RAAN(t_o) + \dot{\Omega}(t - t_o)$$

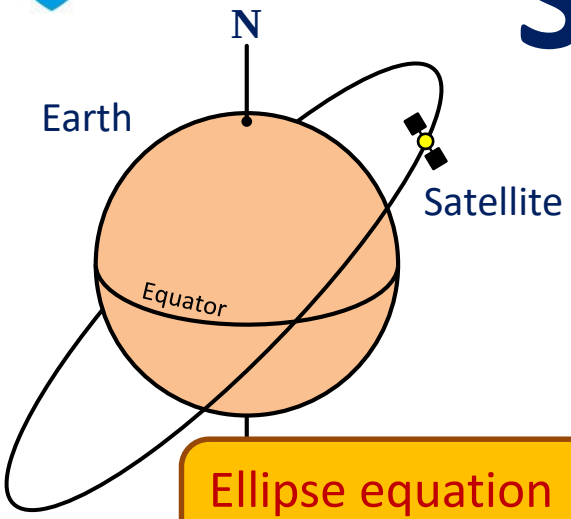
Rate of right ascension, for GPS satellites:

$$\dot{\Omega} \approx -0.04 \, ^{\circ}/\mathrm{day}$$ (nodal precession)

Longitude of the Ascending Node

$$\Omega(t) = RAAN(t) - \Theta(t)$$

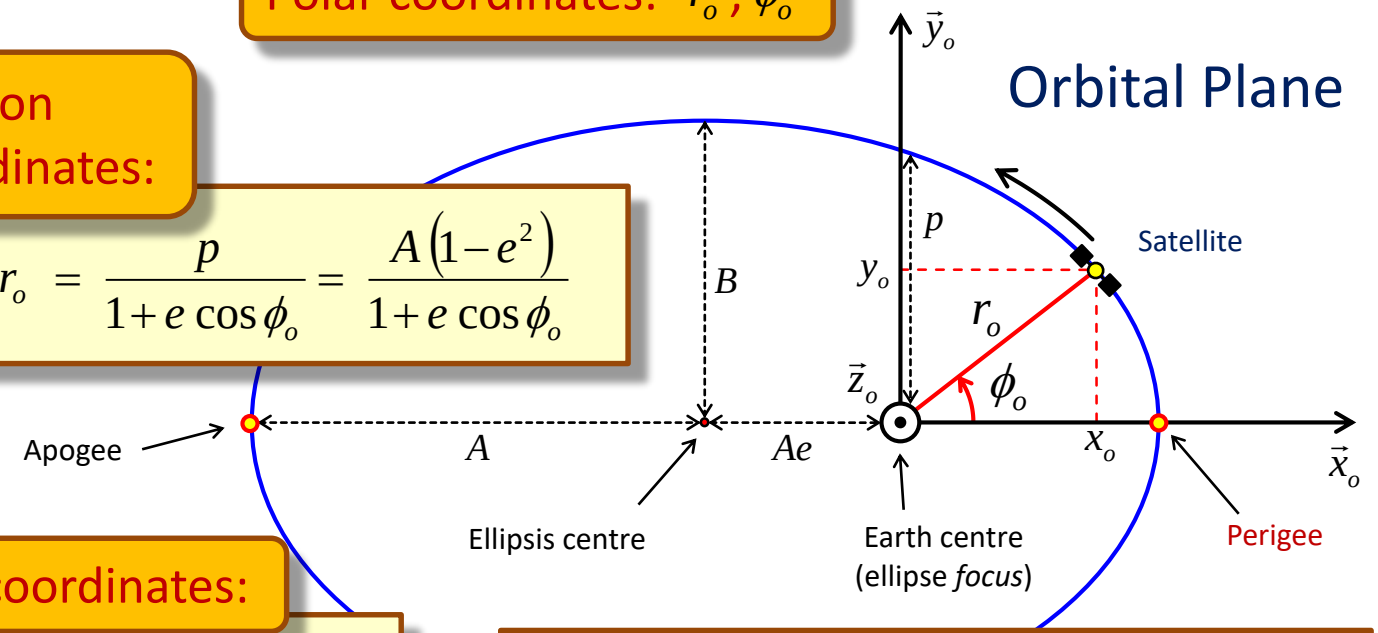
Satellite Coordinates in the Orbital Plane



Polar coordinates: r_o , ϕ_o

Ellipse equation in polar coordinates:

$$r_o = \frac{p}{1+e\cos\phi_o} = \frac{A\left(1-e^2\right)}{1+e\cos\phi_o}$$



Cartesian coordinates:

$$x_o = r_o \cos\phi_o$$
$$y_o = r_o \sin\phi_o$$
$$z_o = 0$$

A – semi-major axis of the orbital ellipse
B – semi-minor axis of the orbital ellipse
e – ellipse eccentricity
p – ellipse *semilatus rectum* $p = A\left(1-e^2\right) = B\sqrt{1-e^2}$

Kepler's 1st Law
The orbits of the planets are ellipses with the Sun at one *focus* (1602).

$\phi_o = ?$

Mean Angular Velocity and Mean Anomaly

Mean angular satellite velocity (a.k.a. satellite mean motion):

$$\eta = \frac{2\pi}{T} = \sqrt{\frac{\mu}{A^3}}$$

Earth gravitational constant: $\mu = 3.986005 \times 10^{14}\ m^3 s^{-2}$

(IS GPS 200)

Mean anomaly:

(mathematical abstraction related with the mean angular motion)

$$M(t) = \eta\left(t - t_p\right) = \eta\left(t_{oe} - t_p\right) + \eta\left(t - t_{oe}\right) = M_o + \eta\left(t - t_{oe}\right)$$

M_o – mean anomaly at time t_{oe}

t_{oe} – reference time epoch

t_p – time of perigee passage

Anomalies

of the Keplerian Orbit

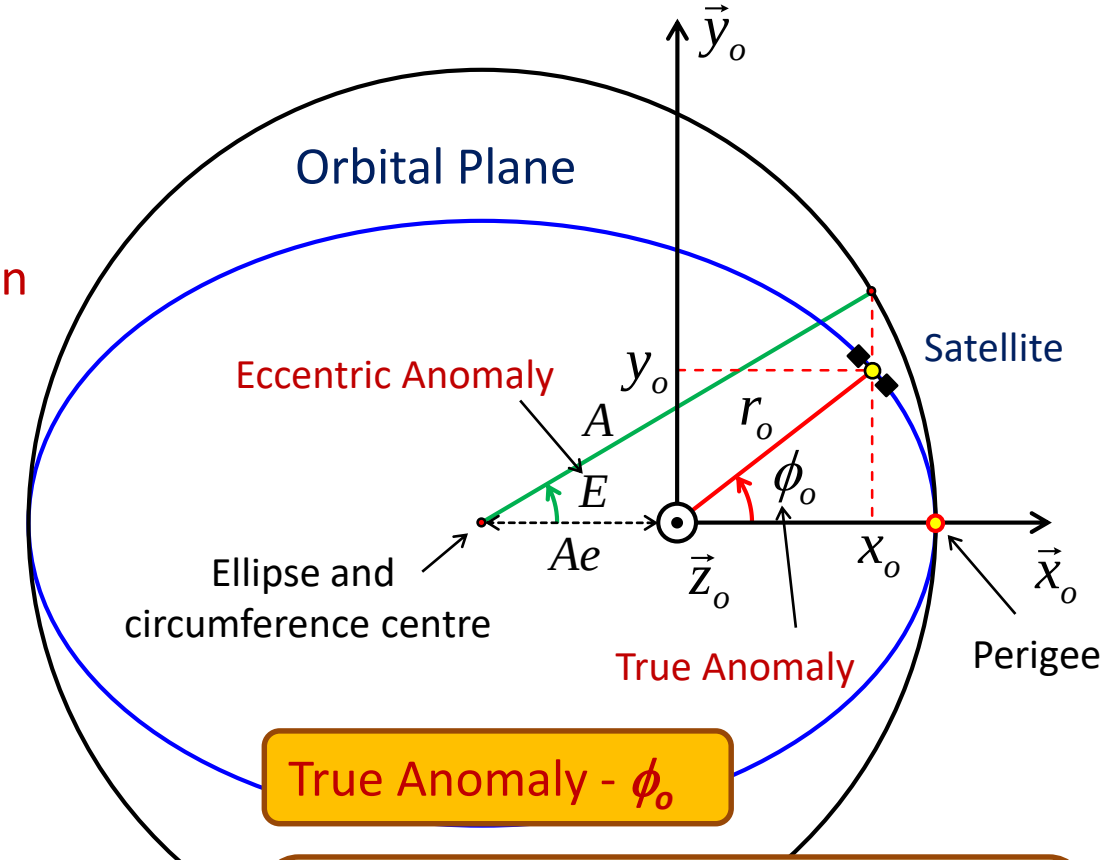
Mean Anomaly - M

$$M(t) = \eta\left(t - t_p\right) = M_o + \eta\left(t - t_{oe}\right)$$

Eccentric Anomaly - E

$$E(t) = M(t) + e\sin E(t)$$

Kepler's Equation



Approach:

$$M(t) \rightarrow E(t) \rightarrow \left[r_o(t), \phi_o(t)\right]$$

Ellipses equation:

$$r_o(t) = \frac{A\left(1-e^2\right)}{1+e\cos\phi_o(t)} \rightarrow \cos\phi_o(t) = \frac{A\left(1-e^2\right)-r_o(t)}{r_o(t)\,e}$$

From the figure:

$$A\cos E(t) = Ae + r_o(t)\cos\phi_o(t)$$
$$x_o(t) = A\left(\cos E(t) - e\right)$$
$$x_o^2(t) + y_o^2(t) = r_o^2(t)$$

$$r_o(t) = A\left(1 - e\cos E(t)\right)$$

$$y_o(t) = \cancel{\pm} A\sqrt{1-e^2}\,\sin E(t)$$

Note that, E and ϕ_o are in the same half plane ([0, π] or [π, 2π]), i.e. sign(y_o) = sign(sin E).

True Anomaly - ϕ_o

$$\phi_o(t) = \arctan\left(\frac{y_o(t)}{x_o(t)}\right) = \arctan\left(\frac{\sqrt{1-e^2}\,\sin E(t)}{\cos E(t) - e}\right)$$

MATLAB hint: use `atan2` to compute ϕ_o

True Anomaly and Tangent Half Angle Formulas

True Anomaly: $\phi_o = \arctan(\frac{y_o}{x_o})$

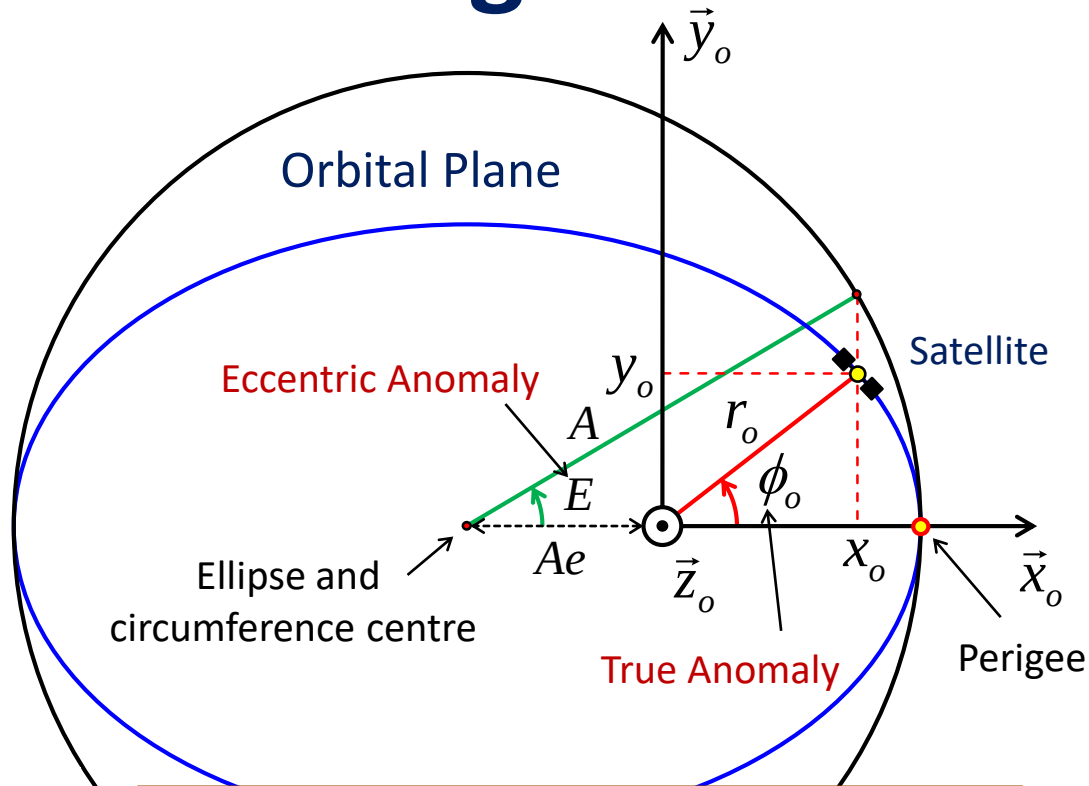
Problem: Range of arctan is $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

Solution:

$$\phi_o = 2\arctan(\tan\frac{\phi_o}{2})$$

Range: $[-\pi, \pi]$ ✓

$$\tan\frac{\phi_o}{2} = \frac{\sin\phi_o}{1+\cos\phi_o}$$



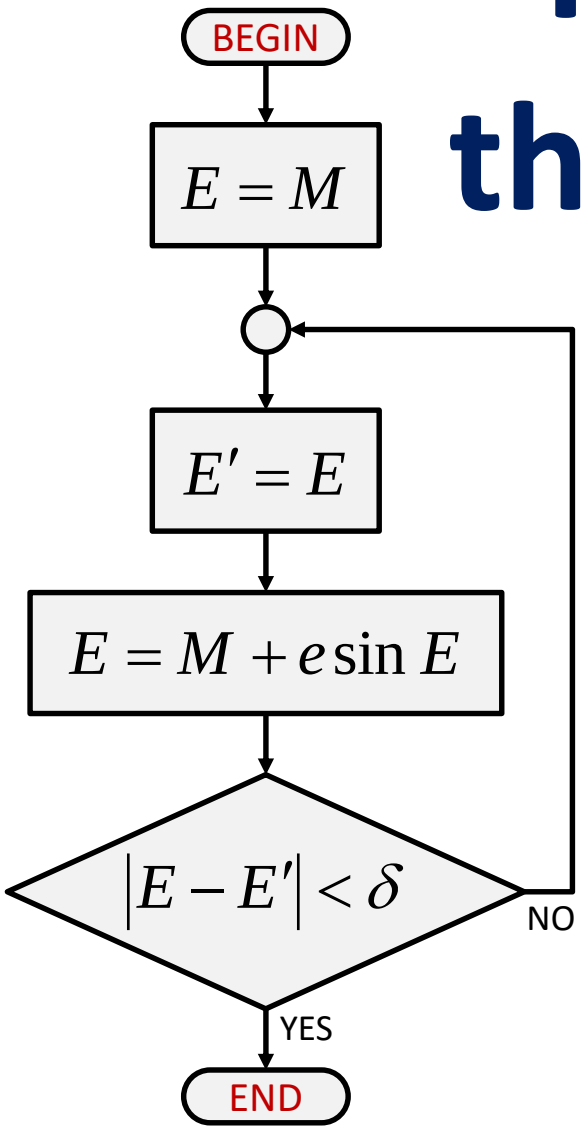
$$\tan\frac{\phi_o}{2} = \frac{\sin\phi_o}{1+\cos\phi_o} = \frac{y_o/r_o}{1+x_o/r_o} = \frac{A\sqrt{1-e^2}\sin E/r_o}{1+(A\cos E - Ae)/r_o}$$

$$= \frac{A\sqrt{1-e^2}\sin E}{r_o + (A\cos E - Ae)} = \frac{A\sqrt{1-e^2}\sin E}{A(1-e\cos E) + (A\cos E - Ae)}$$

$$= \frac{\sqrt{(1-e)(1+e)}\sin E}{(1-e)(1+\cos E)} = \sqrt{\frac{1+e}{1-e}}\tan\frac{E}{2}$$

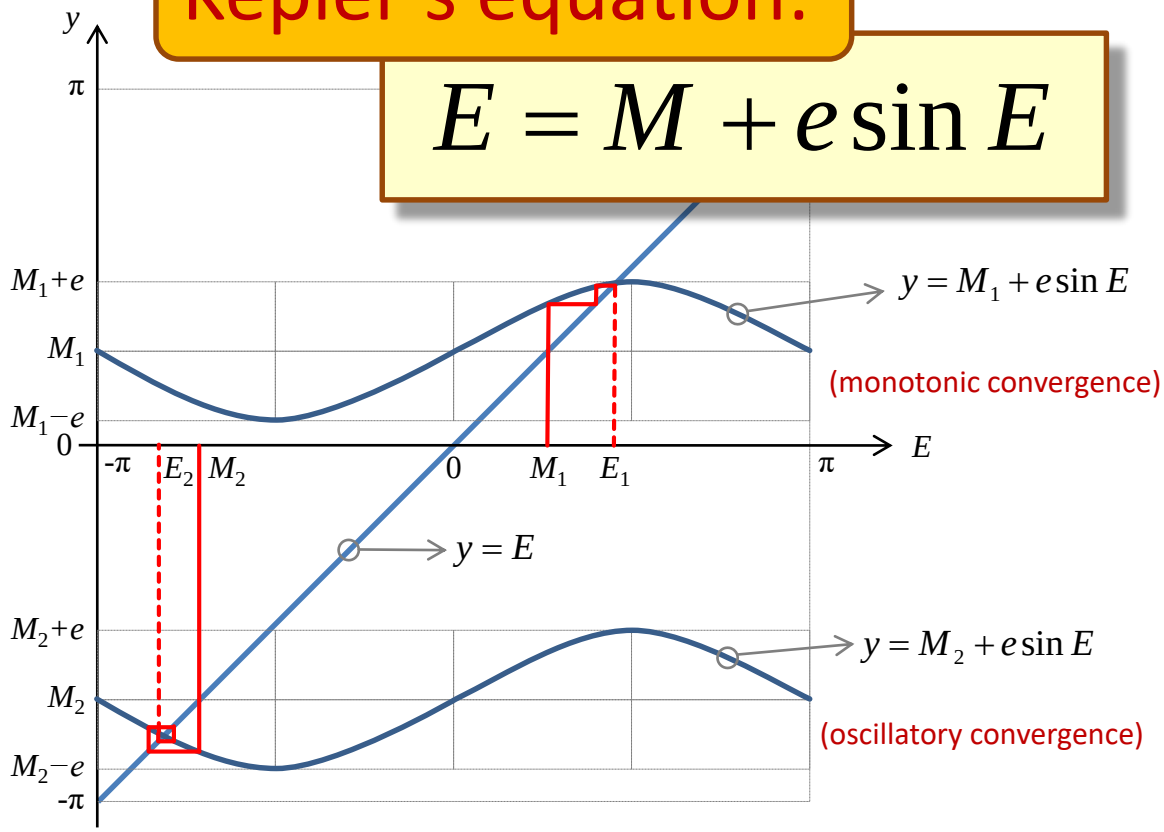
$$\phi_o = 2\arctan\left(\sqrt{\frac{1+e}{1-e}}\tan\frac{E}{2}\right)$$

Iterative Solution of the Kepler's Equation



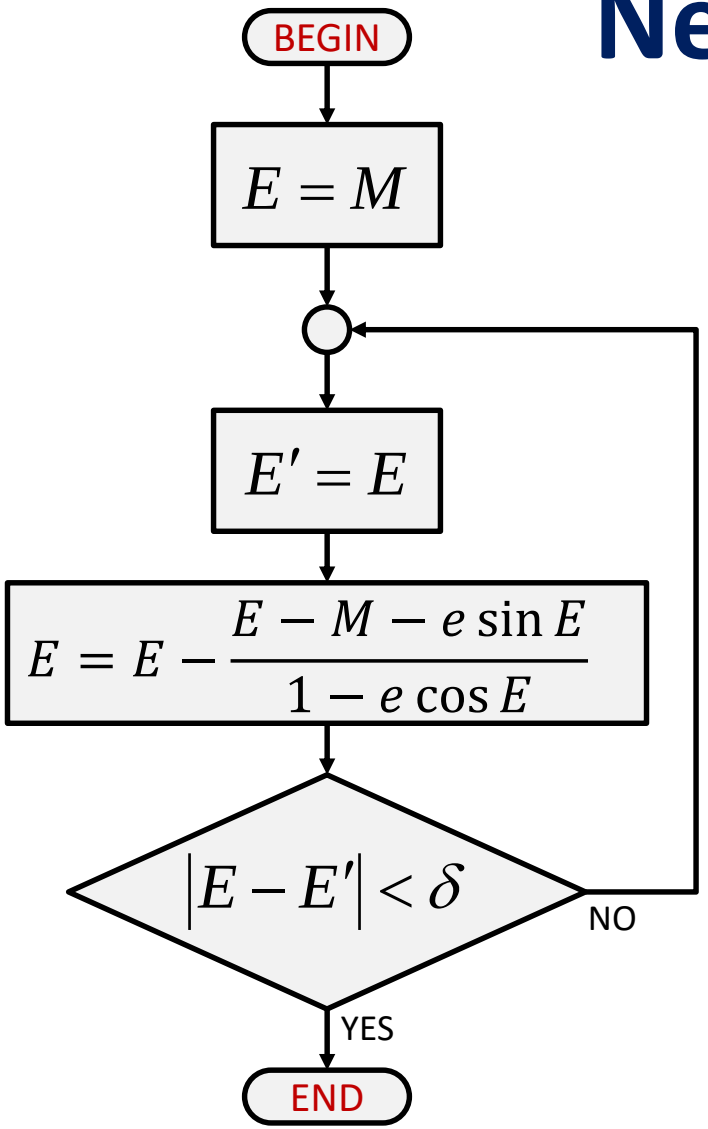
Kepler's equation:

$$E = M + e\sin E$$



The Radio Technical Commission for Maritime Services (RTCM) Special Committee (SC) 104 (Differential Global Navigation Satellite Systems) suggests using δ equal to 1e-12 rad (v2.1 p.III-1).

Iterative Solution based on the Newton-Raphson method



Kepler's equation:

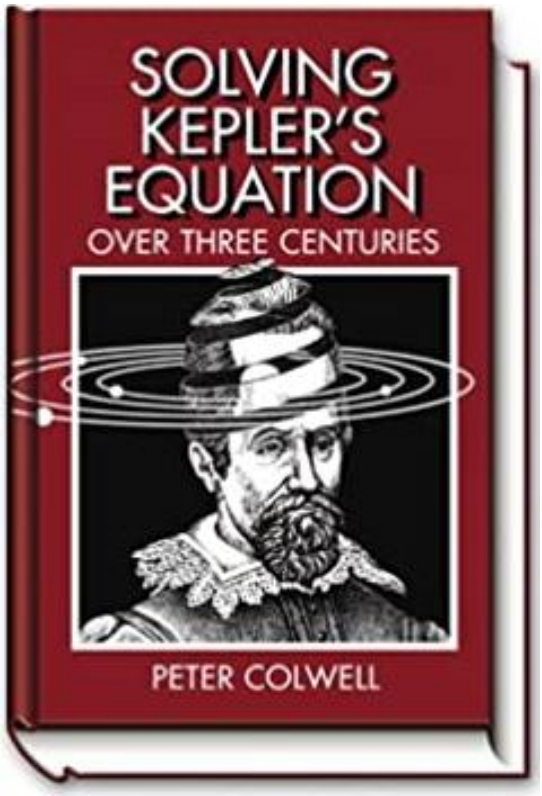
$$E = M + e\sin E$$

Newton-Raphson method:

$$f(E) = E - (M + e\sin E)$$

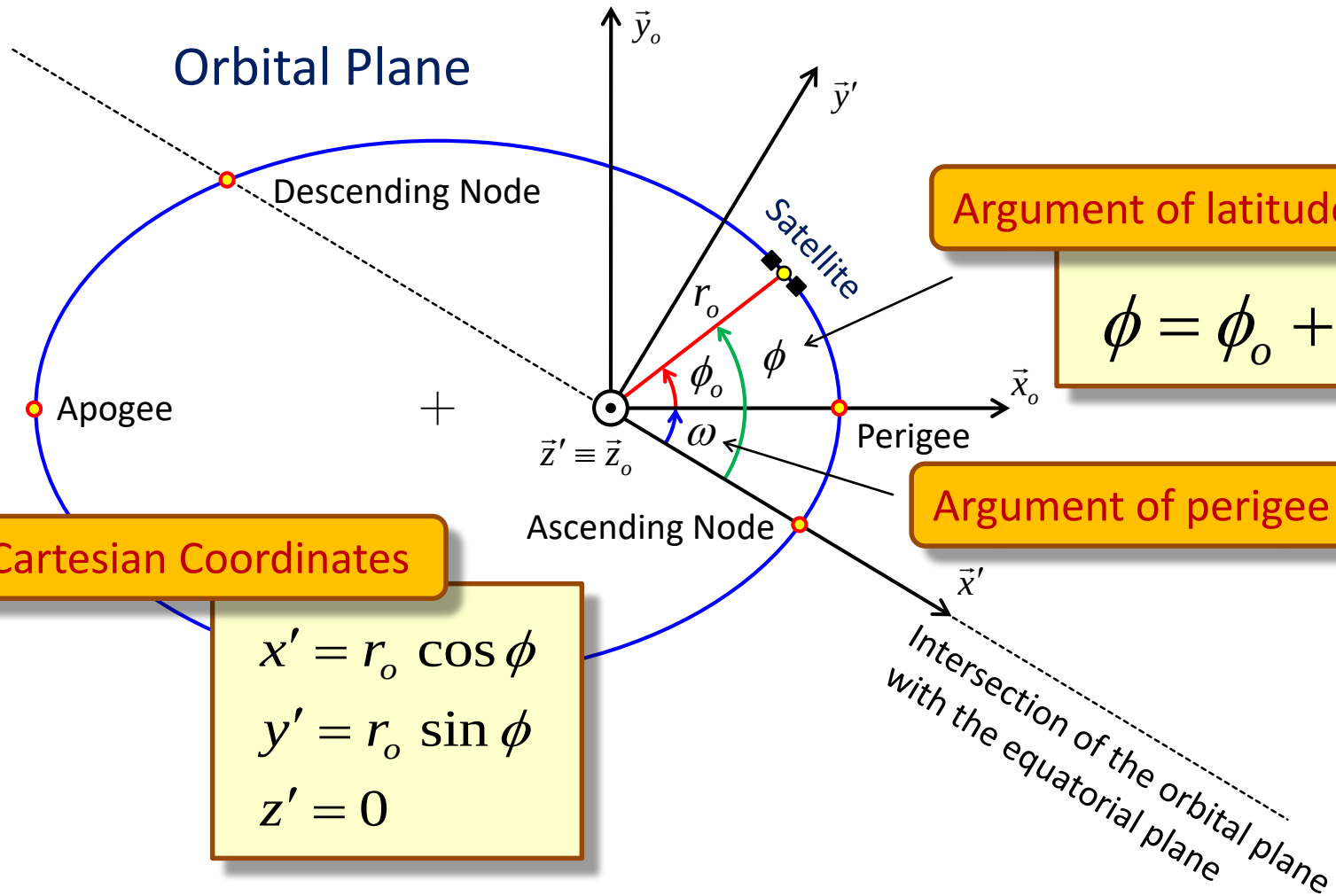
$$E_{n+1} = E_n - \frac{f(E_n)}{f'(E_n)} = E_n - \frac{E_n - M - e\sin E_n}{1 - e\cos E_n}$$

more on this subject…



Peter Colwell,
Solving Kepler's Equation over Three Centuries,
Willmann-Bell, Richmond, 1993.

Argument of Perigee and Argument of Latitude



Argument of latitude - ϕ

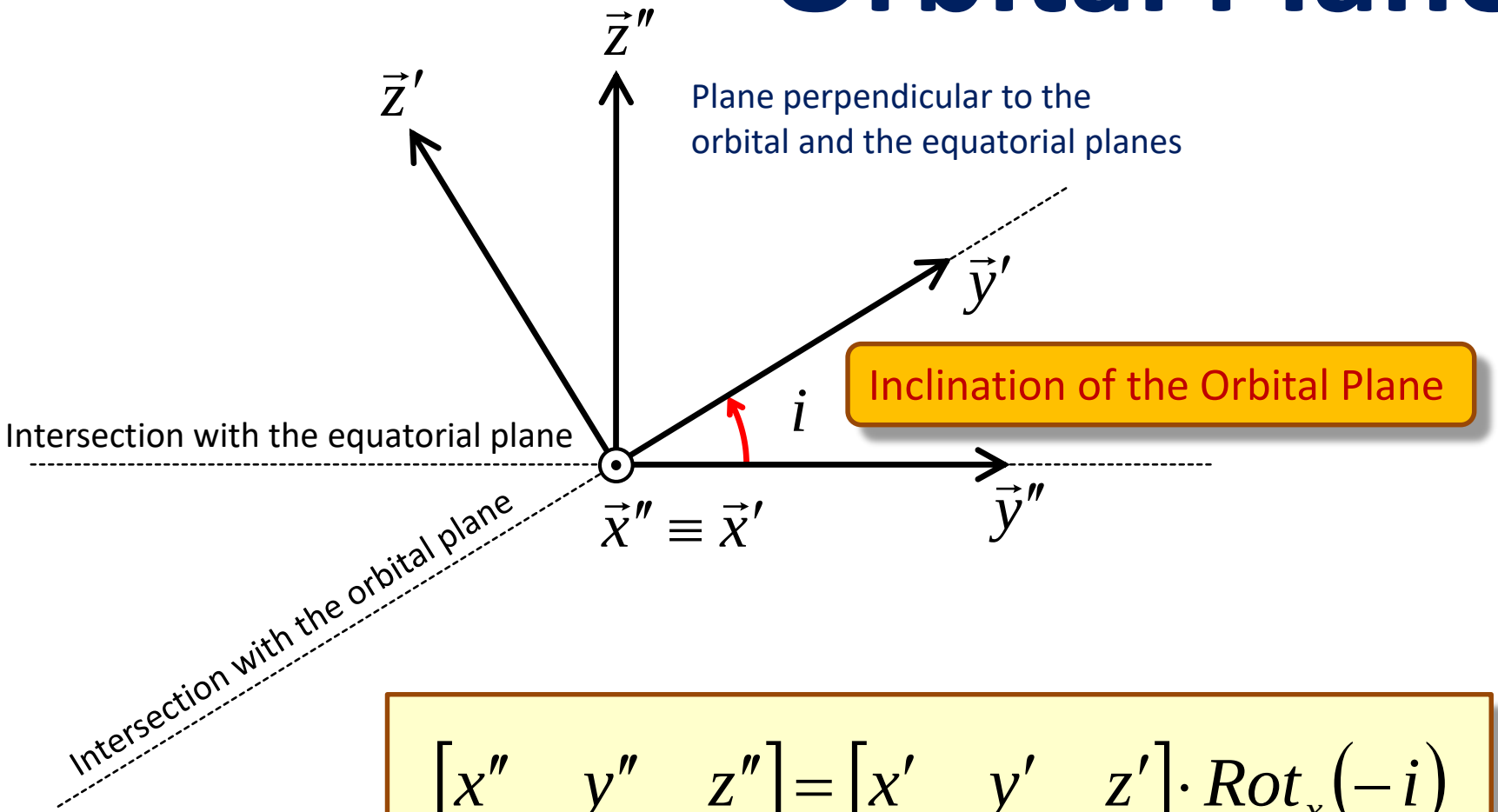
$$\phi = \phi_o + \omega$$

Argument of perigee - ω

Cartesian Coordinates

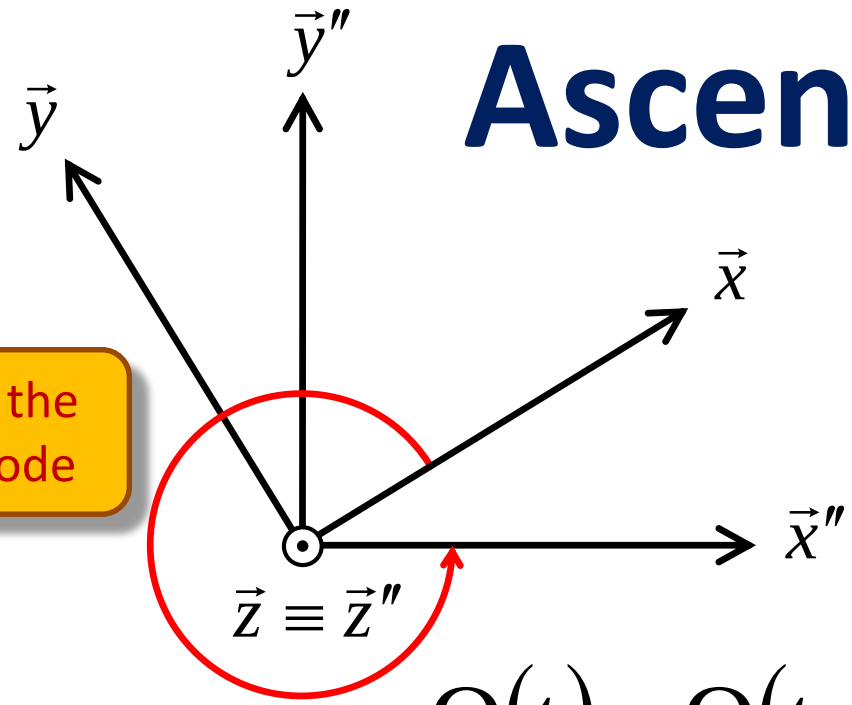
$$x' = r_o \cos\phi$$
$$y' = r_o \sin\phi$$
$$z' = 0$$

Inclination of the Orbital Plane



$$\begin{bmatrix} x'' & y'' & z'' \end{bmatrix} = \begin{bmatrix} x' & y' & z' \end{bmatrix} \cdot Rot_x(-i)$$

Longitude of the Ascending Node



Longitude of the Ascending Node

Equatorial Plane

$$\Omega(t) = \Omega(t_o) - \dot{\Omega}_e \, (t - t_o)$$

(disregarding nodal precession)

$$\begin{bmatrix} x & y & z \end{bmatrix} = \begin{bmatrix} x'' & y'' & z'' \end{bmatrix} \cdot Rot_z(-\Omega)$$

Earth rotation rate: $\dot{\Omega}_e = 7.2921151467 \times 10^{-5} \, \mathrm{rad\,s^{-1}}$

Perturbations due to the oblateness of Earth (J_2)

Regression of the node in the equatorial plane

$$\dot{\Omega} = -\frac{3}{2}\eta a_E^2 \frac{\cos i}{A^2(1-e^2)^2} J_2$$

oblateness term

Rotation of the pedigree

$$\dot{\omega} = \frac{3}{4}\eta a_E^2 \frac{5(\cos i)^2 - 1}{A^2(1-e^2)^2} J_2$$

$(\dot{\omega} = 0 \text{ for } i = \arccos\sqrt{1/5} = 63.4°)$

Variation of the mean anomaly

$$\dot{M} = \eta + \dot{m}$$

$$\dot{m} = \frac{3}{4}\eta a_E^2 \frac{3(\cos i)^2 - 1}{A^2(1-e^2)^3} J_2$$

$(\dot{m} = 0 \text{ for } i = \arccos\sqrt{1/3} = 54.7°)$

$a_E = a_{WGS\ 84} = 6378137.0$ m

$J_2 = 1082.63\text{e-}6$ (WGS 84)

Hofmann-Wellenhof B., Lichtenegger H., Collins J. (1994)
Satellite orbits. In: Global Positioning System. Springer, Vienna.

GPS Satellite Ephemeris

$\sqrt{A}$ - Square root of the orbit semi-major axis.

t_{oe} - Ephemeris reference time.

WN - Ephemeris reference week number.

M_o - Mean anomaly at the ephemeris reference time. It represents the circumference arc hypothetically travelled by the satellite between the last perigee passage and the ephemeris reference time (if it was travelling on a circumference of radius A, with a constant angular velocity, equal to the mean angular velocity).

e - Orbit eccentricity.

i_0 - Orbit inclination angle (relatively to the equator) at the ephemeris reference time.

Ω_0 - Longitude (measured East) of ascending node at weekly epoch (beginning of the week).

ω - Argument of perigee. It represents the angle, measured in the orbital plane, between the ascending node and the perigee.

GPS Satellite Ephemeris (cont.)

$\Delta\eta$ - Correction term to the satellite mean angular velocity.

$\dot{\Omega}$ - Correction term to the right ascension of the ascending node.

C_{uc}, C_{us} - Amplitude of the cosine and sine harmonic correction terms to the argument of latitude.

C_{rc}, C_{rs} - Amplitude of the cosine and sine harmonic correction terms to the orbit radius.

C_{ic}, C_{is} - Amplitude of the cosine and sine harmonic correction terms to the orbit inclination angle.

$IDOT$ - Rate of the orbit inclination angle.

$IODE$ - (*Issue of Data, Ephemeris*) Counter incremented with each new ephemeris dataset.

Almanac

Coarse orbit and status information for each satellite in the constellation

WN_a - Almanac reference week number.

t_{oa} - Almanac reference time.

M_o - Mean anomaly at the almanac reference time.

$\sqrt{A}$ - Square root of the orbit semi-major axis.

e - Orbit eccentricity.

Ω_0 - Longitude (measured East) of ascending node at weekly epoch (beginning of the week).

δ_i - Correction term to the orbit inclination angle, relatively to i_0 = 0.30 sc (sc: semicírculos).

ω - Argument of perigee.

$\dot{\Omega}$ - Correction term to the right ascension of the ascending node.

GPS SV Position (Overview)

1. Orbit semi-major axis: $$A = \left(\sqrt{A}\right)^2$$

2. Corrected satellite mean angular velocity: $$\eta = \sqrt{\frac{\mu}{A^3}} + \Delta\eta$$

3. Time difference between t and t_{oe}: $$\Delta t = t - t_{oe}$$

4. Mean anomaly: $$M = M_o + \eta\,\Delta t$$

5. Eccentric anomaly (iterative method): $$E = M + e\sin E$$

6. True anomaly: $$\phi_o = 2\arctan\left(\sqrt{(1+e)/(1-e)}\tan\frac{E}{2}\right)$$

7. Argument of latitude: $$\phi = \phi_o + \omega$$

8. Corrected argument of latitude: $$u = \phi + \delta_u$$
$$\delta_u = C_{uc}\cos 2\phi + C_{us}\sin 2\phi$$

Overview (cont.)

9. Orbital radius:

$$r_o = A\left(1 - e\cos E\right)$$

10. Corrected orbit radius:

$$r = r_o + \delta_r$$

$$\delta_r = C_{rc}\cos 2\phi + C_{rs}\sin 2\phi$$

11. Corrected angle of inclination:

$$i = i_o + \delta_i + \left(IDOT\right)\Delta t$$

$$\delta_i = C_{ic}\cos 2\phi + C_{is}\sin 2\phi$$

12. Corrected longitude of the ascending node:

$$\Omega = \Omega_o + \left(\dot{\Omega} - \dot{\Omega}_e\right)\Delta t - \dot{\Omega}_e\, t_{oe}$$

13. WGS 84 satellite Cartesian coordinates, at time t:

$$\begin{bmatrix} x & y & z \end{bmatrix} = \begin{bmatrix} r\cos u & r\sin u & 0 \end{bmatrix} \cdot Rot_x\left(-i\right) \cdot Rot_z\left(-\Omega\right)$$

Computing Δt

$$\Delta t = t - t_{oe}$$

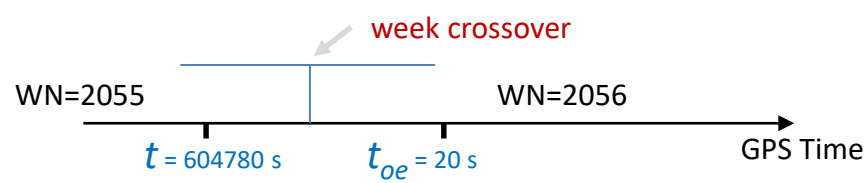
time instant at which the satellite position is to be computed

t_{oe} is included in the Navigation Message (as well as the corresponding WN)

Δt shall be the actual total time difference between the time t and the epoch time t_{oe}, and must account for beginning or end of week crossovers.

Each time instance, in the GPS time scale, is defined by a week number, WN (week number 0 started at 0h UTC – midnight – of January 5th to 6th, 1980), and a time of week, TOW (number of seconds since the beginning of the week).

Example

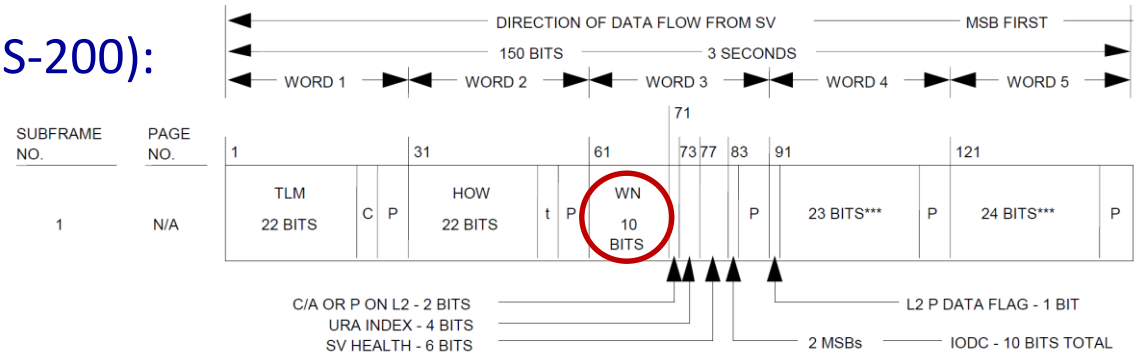


$$\Delta t = t - t_{oe} = -40 \text{ s}$$

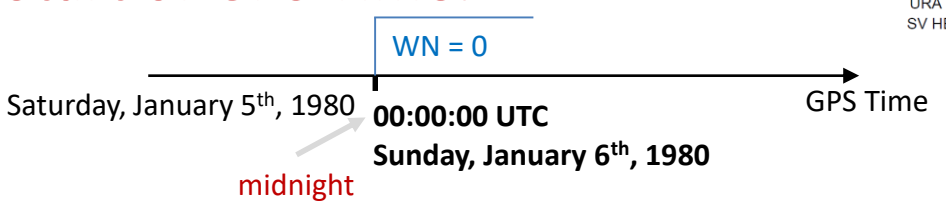
One week has 7×24×60×60 = 604800 seconds

GPS WN Rollover

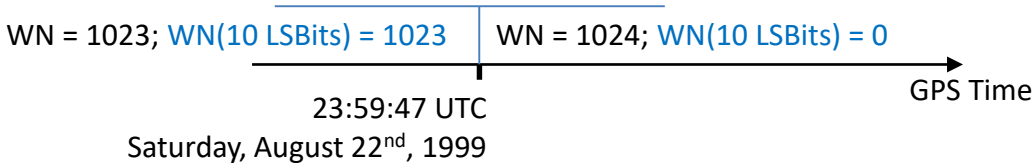
Navigation message (IS-GPS-200):



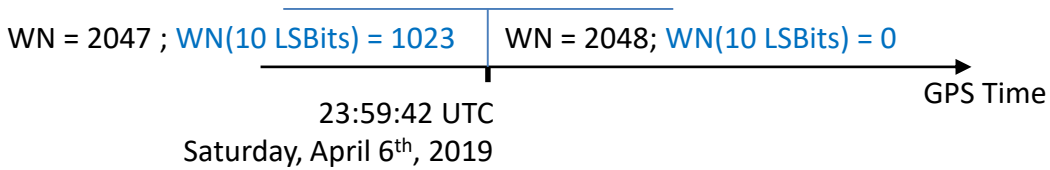
Start of GPS Time:



First Rollover:



Second Rollover:



UTC Leap Seconds

(since 1980)

Year	Jun 30	Dec 31
1980	0	0
1981	+1	0
1982	+1	0
1983	+1	0
1984	0	0
1985	+1	0
1986	0	0
1987	0	+1
1988	0	0
1989	0	+1
1990	0	+1
1991	0	0
1992	+1	0
1993	+1	0
1994	+1	0
1995	0	+1
1996	0	0
1997	+1	0
1998	0	+1
1999	0	0
2000	0	0
2001	0	0
2002	0	0
2003	0	0
2004	0	0
2005	0	+1
2006	0	0
2007	0	0
2008	0	+1
2009	0	0
2010	0	0
2011	0	0
2012	+1	0
2013	0	0
2014	0	0
2015	+1	0
2016	0	+1
2017	0	0
2018	0	0
2019	0	0
2020	0	0
2021	0	0
2022	0	-

23:59:59 December 31, 2005
23:59:60 December 31, 2005
00:00:00 January 1, 2006

Coordinated Universal Time (UTC)
International Earth Rotation and Reference Systems Service (IERS)

(more on) Longitude of Ascending Node (Ω)

(True if drift rate was zero!)

Ω_o - Longitude of Ascending Node of Orbit Plane at Weekly Epoch
(IS-GPS-200 definition)

$t = 0$

Corrected Longitude of Ascending Node:

$$\Omega = \Omega_o - \dot{\Omega}_e t_{oe} + \left(\dot{\Omega} - \dot{\Omega}_e\right)\Delta t$$

with $\Delta t = t - t_{oe}$

drift rate of the ascending node
(due to the nodal precession)

$$t = t_{oe} \longrightarrow \Omega = \Omega_o - \dot{\Omega}_e t_{oe}$$

Longitude of Ascending Node of Orbit Plane at t_{oe}.

$$t = 0 \longrightarrow \Omega = \Omega_o - \dot{\Omega} t_{oe}$$

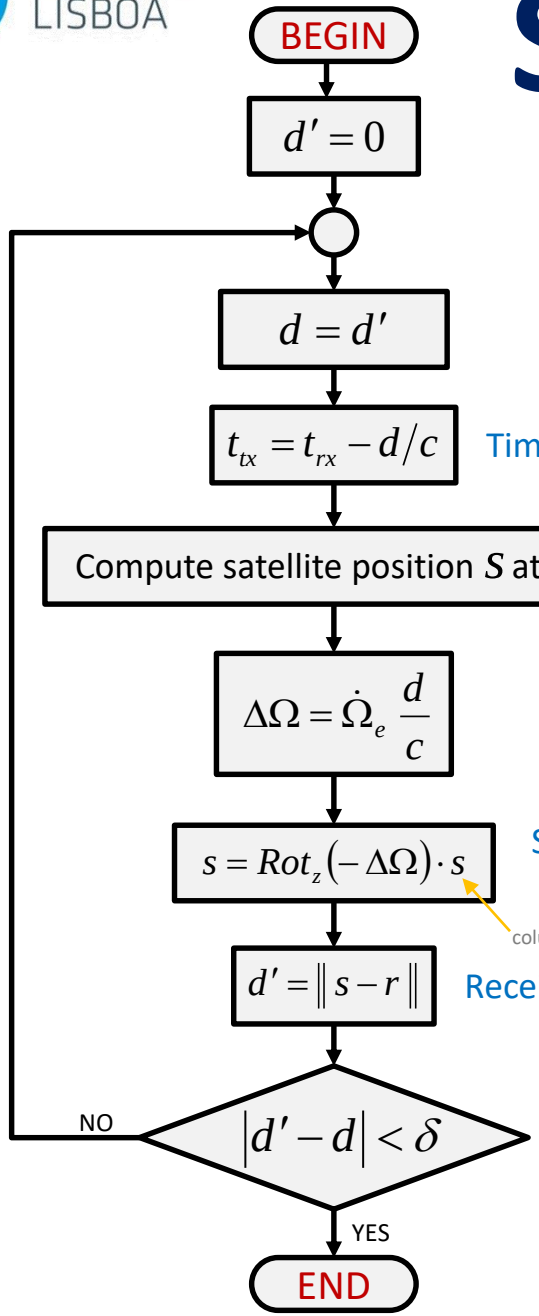
True Longitude of Ascending Node of Orbit Plane at Weekly Epoch.

weekly epoch

Earth rotation rate: $\dot{\Omega}_e = 7.2921151467 \times 10^{-5}\,\mathrm{rad\,s^{-1}}$

Satellite Position @ Time of Transmission

The time of transmission is unknown!



Time of transmission

Use the satellite ephemeris

Earth rotation during the signal travel time

Satellite coordinates in the receiver reference frame

column vector

Receiver to satellite range

Use δ = 1mm

If s is a row vector

$s = s \cdot Rot_z(\Delta\Omega)$

row vector